

Shrinking the Cross Section

Theory, Methodology, and Replication Results

Based on Kozak, Nagel, and Santosh (2018)

Financial Econometrics, Honor

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Big Question

The factor zoo problem

Empirical asset pricing has found many firm characteristics that seem to predict the cross-section of stock returns.

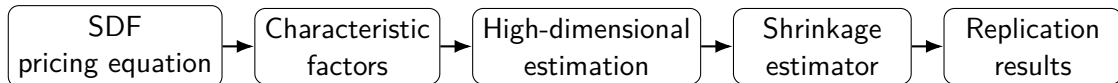
- Traditional models try to summarize expected returns with a few factors.
- But the candidate set is now large: value, momentum, profitability, investment, volatility, liquidity, and many financial ratios.
- The paper asks how to estimate a stochastic discount factor when the number of candidate factors is large.

Main idea

Do not force the SDF to be sparse in raw characteristics. Use economically motivated shrinkage to control overfitting.

Roadmap

- ① What is a stochastic discount factor (SDF)?
- ② How does an SDF connect to factor models?
- ③ What is the traditional estimation approach?
- ④ Why does it fail in a high-dimensional cross-section?
- ⑤ What method does this paper propose?
- ⑥ What do our replication results show?



SDF: The Object That Prices Payoffs

Consider a one-period payoff X_{t+1} . Its price today is P_t .

Basic pricing formula

$$P_t = \mathbb{E}_t [M_{t+1} X_{t+1}].$$

- M_{t+1} is the stochastic discount factor.
- It is stochastic because the value of money tomorrow depends on the state of the world.
- It is a discount factor because it converts future payoffs into today's price.

Intuition

The SDF tells us how valuable one unit of payoff is in each future state.

Why Is the SDF State-Dependent?

A payoff is more valuable when it arrives in states where investors value wealth more.

State tomorrow	Consumption	Marginal value of wealth	SDF M
Bad state	Low	High	High
Good state	High	Low	Low

- A payoff in bad states provides insurance, so it is expensive.
- A payoff in good states provides less insurance, so it is cheaper.
- Therefore, expected return is not enough. Timing across states matters.

A Two-State Example

Suppose an asset pays X_B in a bad state and X_G in a good state.

$$P = p_B M_B X_B + p_G M_G X_G, \quad M_B > M_G.$$

Insurance-like asset

X_B high, X_G low

The payoff is high when M is high. Investors value it highly, so its price is high and expected return is low.

Procyclical asset

X_B low, X_G high

The payoff is high when M is low. Investors require a higher expected return to hold it.

From Payoffs to Returns

If R_{t+1} is the gross return of an asset, then $X_{t+1} = P_t R_{t+1}$.

Substitute into the pricing formula:

$$P_t = \mathbb{E}_t[M_{t+1} P_t R_{t+1}].$$

Divide both sides by P_t :

$$\mathbb{E}_t[M_{t+1} R_{t+1}] = 1.$$

For an excess return R_{t+1}^e , the price is zero:

$$\mathbb{E}_t[M_{t+1} R_{t+1}^e] = 0.$$

This is the key asset pricing equation

In this paper, the SDF is estimated so that this equation holds as well as possible for many factor returns.

Risk Premia Through the SDF

Start from the excess return condition:

$$\mathbb{E}[MR^e] = 0.$$

Use $\mathbb{E}[MR^e] = \mathbb{E}[M]\mathbb{E}[R^e] + \text{Cov}(M, R^e)$:

$$\mathbb{E}[R^e] = -\frac{\text{Cov}(M, R^e)}{\mathbb{E}[M]}.$$

- If an asset pays off when M is high, then $\text{Cov}(M, R^e) > 0$.
- It is a hedge against bad times, so it earns a low expected return.
- If an asset pays off when M is low, then $\text{Cov}(M, R^e) < 0$.
- It performs poorly in valuable states, so it must earn a high expected return.

Linear SDF and Factor Models

Empirical asset pricing often assumes the SDF is linear in factor returns:

$$M_t = 1 - \mathbf{b}'(\mathbf{F}_t - \mathbb{E}[\mathbf{F}_t]).$$

Here:

- $\mathbf{F}_t$ is a vector of factor returns.
- $\mathbf{b}$ is the vector of SDF loadings.
- A factor receives a large loading when it is important for pricing assets.

Plugging the linear SDF into the pricing equation gives:

$$\mathbb{E}[M_t \mathbf{F}_t] = \mathbf{0}.$$

Estimation target

Estimate $\mathbf{b}$, the SDF weights on candidate factor returns.

From Characteristics to Factor Returns

Let:

- R_t : vector of individual stock excess returns.
- Z_{t-1} : matrix of lagged firm characteristics.
- $F_t = Z'_{t-1} R_t$: characteristic-based factor returns.

Examples of characteristics:

Size	Book-to-market	Momentum	Profitability
Investment	Volatility	Liquidity	Financial ratios

Interpretation

A characteristic-based factor is a long-short portfolio constructed from a firm characteristic.

Known Moments: The Ideal Estimator

Define:

$$\boldsymbol{\mu} = \mathbb{E}[\mathbf{F}_t], \quad \boldsymbol{\Sigma} = \text{Var}(\mathbf{F}_t).$$

With the linear SDF

$$M_t = 1 - \mathbf{b}'(\mathbf{F}_t - \boldsymbol{\mu}),$$

the pricing equation implies:

$$\mathbf{b} = \boldsymbol{\Sigma}^{-1}\boldsymbol{\mu}.$$

Important interpretation

The SDF weights are the weights of a mean-variance efficient portfolio formed from the factor returns.

- High expected return matters.
- But covariance with other factors also matters.

Naive Sample Estimation

In practice, population moments are unknown:

$$\mu \rightarrow \hat{\mu}, \quad \Sigma \rightarrow \hat{\Sigma}.$$

The naive estimator is:

$$\hat{\mathbf{b}}_{\text{OLS}} = \hat{\Sigma}^{-1} \hat{\mu}.$$

This is similar to running a cross-sectional regression of average returns on covariances.

Main difficulty

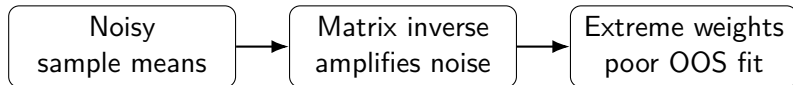
Average returns are very noisy, and the noise becomes dangerous when the number of candidate factors is large.

High Dimensionality and Overfitting

Suppose we have H candidate factors and T time-series observations.

When H is large:

- many sample means look large just by chance;
- the covariance matrix is harder to invert reliably;
- the estimator can assign extreme weights to noisy directions;
- in-sample fit may look good, but out-of-sample performance is poor.



Why Not Just Use a Few Factors?

A common strategy is to impose a sparse model:

$$M_t = 1 - b_1 F_{1t} - \cdots - b_K F_{Kt}, \quad K \text{ small.}$$

Examples include three-factor, five-factor, or six-factor models.

The paper calls this a **characteristics-sparse SDF**.

The authors' concern

There is no strong economic reason why only a few observable characteristics should summarize all cross-sectional expected returns.

- Economic models involve unobservable objects such as expected profitability and planned investment.
- Many observed characteristics may proxy for these objects.
- Therefore, sparsity in raw characteristics should be tested, not assumed.

Shrinkage Estimator

The paper starts with an L_2 -type shrinkage estimator:

$$\hat{\mathbf{b}} = (\mathbf{\Sigma} + \gamma \mathbf{I})^{-1} \hat{\boldsymbol{\mu}}.$$

Compare it with the naive estimator:

$$\hat{\mathbf{b}}_{\text{OLS}} = \mathbf{\Sigma}^{-1} \hat{\boldsymbol{\mu}}.$$

- $\gamma > 0$ controls the strength of shrinkage.
- Larger γ pushes SDF weights closer to zero.
- Shrinkage sacrifices some in-sample fit to improve out-of-sample stability.

Core idea

Keep many candidate factors, but prevent noisy directions from receiving implausibly large SDF weights.

Bayesian Motivation

The paper motivates shrinkage using a prior on factor expected returns:

$$\boldsymbol{\mu} \sim N\left(\mathbf{0}, \frac{\kappa^2}{\tau} \boldsymbol{\Sigma}^\eta\right).$$

The key choice is:

$$\eta = 2.$$

- The prior connects first moments to second moments.
- Large expected returns should be associated with important sources of return variation.
- Very high Sharpe ratios in low-variance directions are economically implausible.

Economic interpretation

The prior rules out near-arbitrage opportunities as the default explanation for high sample average returns.

Principal Component Interpretation

Decompose the covariance matrix:

$$\Sigma = \mathbf{Q}\mathbf{D}\mathbf{Q}',$$

where $\mathbf{D}$ contains eigenvalues $d_1, \dots, d_H$.

Define principal component factor returns:

$$\mathbf{P}_t = \mathbf{Q}'\mathbf{F}_t.$$

In PC space, the shrinkage estimator has component:

$$\hat{b}_{P,j} = \left(\frac{d_j}{d_j + \gamma} \right) \left(\frac{\hat{\mu}_{P,j}}{d_j} \right).$$

Key mechanism

Low-eigenvalue PCs receive stronger shrinkage because $\frac{d_j}{d_j + \gamma}$ is smaller when d_j is small.

Why Shrink Low-Variance PCs More?

A low-variance PC with a high average return has an extremely high Sharpe ratio.

$$\text{Sharpe ratio} \approx \frac{\text{mean return}}{\text{volatility}}.$$

- If volatility is tiny but average return is large, the Sharpe ratio explodes.
- This resembles a near-arbitrage opportunity.
- Such opportunities are unlikely to survive in well-functioning markets.
- Therefore, high sample means in low-variance PCs are more likely to be noise.

Economic content of the estimator

Major sources of expected return premia should align with major sources of return comovement.

Penalized Estimator View

The same estimator can be written as a penalized optimization problem:

$$\hat{\mathbf{b}} = \arg \min_{\mathbf{b}} \{ (\hat{\boldsymbol{\mu}} - \boldsymbol{\Sigma} \mathbf{b})' (\hat{\boldsymbol{\mu}} - \boldsymbol{\Sigma} \mathbf{b}) + \gamma \mathbf{b}' \boldsymbol{\Sigma} \mathbf{b} \}.$$

Equivalently, it minimizes Hansen-Jagannathan distance with an L_2 penalty:

$$\hat{\mathbf{b}} = \arg \min_{\mathbf{b}} \{ (\hat{\boldsymbol{\mu}} - \boldsymbol{\Sigma} \mathbf{b})' \boldsymbol{\Sigma}^{-1} (\hat{\boldsymbol{\mu}} - \boldsymbol{\Sigma} \mathbf{b}) + \gamma \mathbf{b}' \mathbf{b} \}.$$

- The first term measures pricing errors.
- The second term penalizes extreme SDF weights.

Allowing Sparsity: Adding an L_1 Penalty

The L_2 -only estimator shrinks coefficients, but usually does not set them exactly to zero.

To allow factor selection, the paper adds an L_1 penalty:

$$\hat{\mathbf{b}} = \arg \min_{\mathbf{b}} \left\{ (\hat{\boldsymbol{\mu}} - \boldsymbol{\Sigma} \mathbf{b})' \boldsymbol{\Sigma}^{-1} (\hat{\boldsymbol{\mu}} - \boldsymbol{\Sigma} \mathbf{b}) + \gamma_2 \mathbf{b}' \mathbf{b} + \gamma_1 \sum_{j=1}^H |b_j| \right\}.$$

- L_2 : stabilizes estimation and controls overfitting.
- L_1 : can set some coefficients exactly to zero.
- Together, this resembles elastic net, but with asset-pricing structure.

Why Not Use Lasso Alone?

Pure L_1 methods can be problematic in this setting.

- Characteristic-based factors are highly correlated.
- Lasso tends to select one variable from a group and discard the others.
- But related characteristics may contain a common signal plus different noise.
- Combining them can be better than selecting only one.

Example

Book-to-market, dividend-price, and cash-flow-to-price may all capture a value-related signal. A shrinkage method can average information across them.

The paper's position

Sparsity should be allowed, but not imposed as the main source of robustness.

Choosing the Penalty Strength

The paper chooses penalty parameters by cross-validation.

For each candidate penalty:

- 1 Split the time-series sample into K folds.
- 2 Estimate $\mathbf{b}$ using $K - 1$ folds.
- 3 Evaluate pricing performance on the withheld fold.
- 4 Average the out-of-sample performance across folds.

The criterion is cross-validated out-of-sample R^2 :

$$R_{\text{oos}}^2 = 1 - \frac{(\hat{\mu}_2 - \Sigma_2 \hat{\mathbf{b}})'(\hat{\mu}_2 - \Sigma_2 \hat{\mathbf{b}})}{\hat{\mu}_2' \hat{\mu}_2}.$$

Purpose

Choose shrinkage strength based on out-of-sample pricing performance.

Theory Part: Main Takeaways

- ① The SDF prices assets through

$$\mathbb{E}[MR^e] = 0.$$

- ② In a linear factor model, estimating the SDF means estimating factor weights $\mathbf{b}$.
- ③ The ideal formula $\mathbf{b} = \Sigma^{-1}\boldsymbol{\mu}$ becomes unstable when many candidate factors are used.
- ④ The paper's L_2 shrinkage estimator stabilizes the SDF and has an economic interpretation: it discourages near-arbitrage opportunities.
- ⑤ L_1 can be added to allow sparsity, but the paper treats shrinkage as the central ingredient.

Transition to the Empirical Part

The theory suggests three empirical questions:

- ① Does the shrinkage estimator improve out-of-sample pricing performance?
- ② Is the SDF sparse in raw characteristic-based factors?
- ③ Is the SDF more sparse after rotating factors into principal components?

What the replication will check

The empirical section tests whether economically motivated shrinkage can summarize the pricing information in a large set of stock characteristics.

Original Empirical Design: Core Data Sets

The paper starts with a simple benchmark and then moves to large sets of characteristic-managed portfolios.

- **FF25 size/book-to-market portfolios:** daily returns from 1926–2017. This is a sanity check where a sparse FF-style structure is expected.
- **50 anomaly portfolios:** daily returns from 1973–2017. These are built from published anomaly characteristics and form the main test of characteristics-sparse SDFs.
- **80 WFR portfolios:** daily returns from 1964–2017. These combine WRDS financial ratios with lagged-return portfolios, so the candidate set includes both relevant and potentially irrelevant predictors.

Empirical focus

The key tension is raw-characteristic sparsity versus PC-space sparsity.

Original Empirical Design: Interactions and Powers

The paper also studies extremely high-dimensional versions of the anomaly and WFR data sets.

- The authors add second powers, third powers, and first-order interactions of characteristics.
- This relaxes the simple linear portfolio-weight assumption in characteristic construction.
- The 50-anomaly data expand to 1,375 managed portfolios.
- The 80-WFR data expand to 3,400 managed portfolios.
- Because covariance estimation is harder in this larger space, the paper switches from 3-fold to 2-fold cross-validation.

Purpose

This tests whether shrinkage still works when the candidate SDF basis is far larger than a standard few-factor model.

How the Characteristic Portfolios Are Used

The empirical objects are not individual stocks in the final SDF estimation. They are characteristic-managed long-short portfolio returns.

- For each characteristic, stocks are ranked and transformed into portfolio weights.
- Each factor is a zero-investment long-short portfolio:

$$\mathbf{F}_t = \mathbf{Z}'_{t-1} \mathbf{R}_t.$$

- Portfolio returns are orthogonalized with respect to the market return, so the analysis focuses on cross-sectional pricing information.
- Daily returns are used mainly to estimate covariances more precisely.
- Penalty parameters are chosen by out-of-sample cross-validation.

Why this matters for the slides

The results below evaluate whether the estimated SDF prices these managed portfolios out of sample.

Our Replication Scope

Our replication results focus on the paper's main 50-anomaly exercise.

- Estimation target: the linear SDF

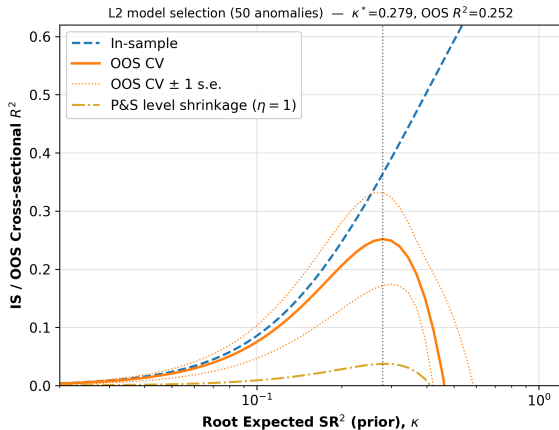
$$M_t = 1 - \mathbf{b}'(\mathbf{F}_t - \mathbb{E}[\mathbf{F}_t]).$$

- Evaluation metric: cross-validated out-of-sample cross-sectional R^2 .
- Main basis assets: 50 anomaly managed portfolios and their 50 principal components.
- Data availability: the delivery package uses portfolio-level returns and public factor data; it does not reconstruct the full WRDS/WFR universe from paid raw stock-accounting data.
- Output: four core replication figures plus one machine-learning extension.

What we want to verify

L_2 shrinkage is essential; raw characteristics are not very sparse; sparsity is much more plausible after rotating factors into principal components.

Result 1: L_2 Shrinkage Selection



Paper link

Figure 4(a), 50 anomaly portfolios.

Key numbers

- OOS R^2 peaks at 0.252.
- Optimal $\kappa^* = 0.279$.
- Level shrinkage peaks at only 0.037.

Axis

Larger κ means weaker L_2 shrinkage.

Result 1: What It Means

The first figure is the cleanest evidence that regularization is necessary.

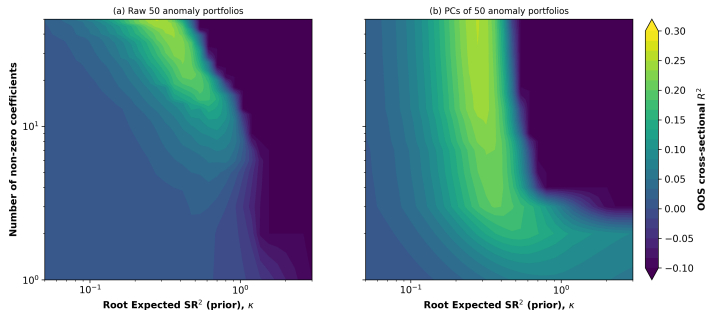
- The dashed in-sample curve keeps improving as shrinkage weakens.
- The OOS curve is hump-shaped: too much shrinkage discards signal, but too little shrinkage overfits noisy sample means.
- The best OOS result occurs at $\kappa^* = 0.279$, not at the unregularized limit.
- The level-shrinkage benchmark performs much worse, which supports the paper's emphasis on relative shrinkage across principal components.

Connection to the theory

The data support the prior intuition: low-variance directions should not be allowed to generate implausibly large SDF weights.

Result 2: Dual Penalty in Raw Factors vs PCs

Fig 2 — Dual-penalty (L1+L2) OOS R^2 (50 anomalies)



Raw characteristics

Corresponds to paper Figure 3(a). With at most 10 nonzero factors, the best OOS R^2 is only 0.108.

Principal components

Corresponds to paper Figure 3(b). With at most 10 nonzero PCs, the best OOS R^2 reaches 0.233.

Result 2: How to Read the Contours

Each panel varies two regularization choices at the same time.

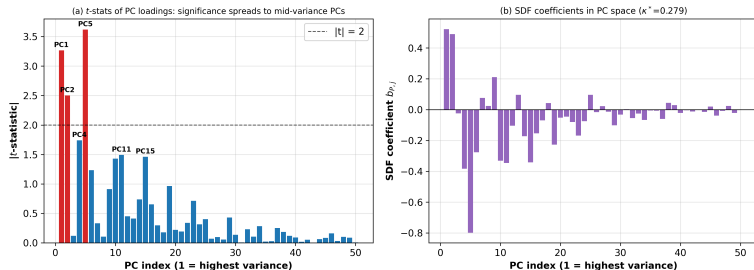
- Horizontal axis: κ , which controls L_2 shrinkage.
- Vertical axis: the number of nonzero SDF coefficients, controlled by the L_1 penalty.
- Color: cross-validated OOS R^2 .
- The raw-characteristic panel reaches high values only when many factors remain nonzero.
- The PC panel keeps high OOS R^2 even with far fewer nonzero factors.

Economic interpretation

The anomalies are not very redundant as raw characteristics, but their covariance structure reveals a lower-dimensional PC representation.

Result 3: SDF Coefficients in PC Space

Fig 3 — SDF coefficient distribution across PCs (50 anomalies)



Paper link

Corresponds to Table 1(b).

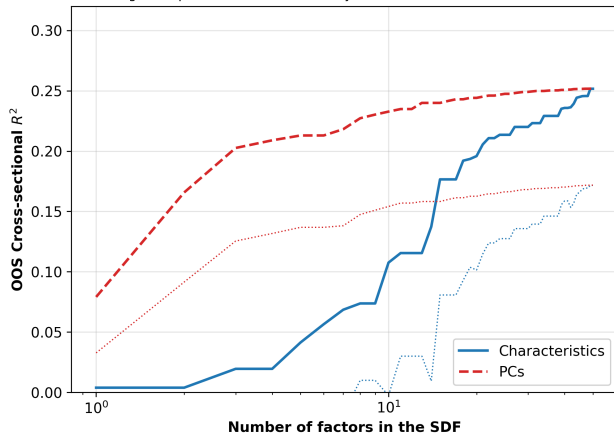
Largest $|t|$ values

PC5, PC1, PC2, PC4, PC11, PC15, PC10.

- The ordering matches the paper's Table 1b closely.
- Pricing information is not concentrated only in PC1.
- Some medium-variance PCs still matter.

Result 4: Sparsity Frontier

Fig 4 — Sparse vs dense: OOS R^2 by number of factors (50 anomalies)



OOS R^2 by sparsity

Factors	Raw char.	PCs
2	0.004	0.166
4	0.020	0.209
10	0.108	0.233
50	0.252	0.252

Takeaway

Characteristic-sparse SDFs perform poorly. PC-sparse SDFs approach the dense model much faster.

Result 4: Link to the Paper's Claim

This figure turns the contour plots into a direct sparsity comparison.

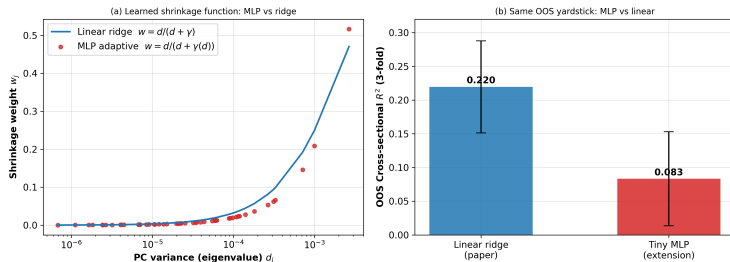
- For each target number of factors, we choose the best L_2 penalty.
- Raw characteristics improve slowly: even 10 factors reach only $R^2 = 0.108$.
- PCs improve quickly: 4 PCs already reach $R^2 = 0.209$, and 10 PCs reach $R^2 = 0.233$.
- The dense 50-factor versions are similar, so the gap is about sparsity, not about the total information set.

Main empirical message

The SDF is not sparse in raw anomaly characteristics, but it can be approximated well by a small number of PCs.

Extension: A Tiny MLP Shrinkage Rule

Fig 5 — ML extension: tiny MLP adaptive shrinkage vs linear ridge (50 anomaly PCs)



Same OOS-CV criterion

MLP : 0.083, ridge : 0.220.

- The MLP learns a nonlinear shrinkage curve.
- Extra flexibility does not improve OOS pricing.
- This supports the paper's view that strong structure is valuable.

Replication Summary

Result	Key number	Interpretation
L_2 selection	OOS $R^2 = 0.252$ at $\kappa^* = 0.279$	Economically motivated shrinkage improves OOS performance.
Dual penalty	PC 0.233 vs raw 0.108, at ≤ 10 factors	Sparsity works much better in PC space.
PC coefficients	PC5, PC1, PC2, PC4	Pricing information is spread across several PCs.
Sparsity frontier	4 PCs give $R^2 = 0.209$, 4 raw factors give 0.020	Few raw characteristics cannot summarize the cross-section.
MLP extension	MLP 0.083 vs ridge 0.220	Flexible nonlinear shrinkage does not beat structured ridge shrinkage.

Overall message

Robust SDF estimation needs shrinkage, and low-dimensional structure is more visible in principal components than in raw characteristics.